

Spacial Mass distributions and Galactic Stability

Advisor: Yuan Shi

Aim

- Warm Up: find a mass distribution for the trajectory:

$$z(t) = \frac{\sqrt{a^2 t^2 + 1} - 1}{a}$$

- Warm Up: Check and see if there are any other solutions for the associated lagrangian:

$$L = -\frac{m_0}{az+1} \sqrt{1 - \dot{z}^2}$$

- Main: Extend the mass distribution idea to a 2-D system and see if the mass distribution can explain Galactic Stability (the issue dark matter solves)

Loschmidt's Reversibility Objection Continued

- I have completed the warm up phase and I am

End